Change-LISA: Language-Guided Reasoning for Remote Sensing Change Detection

Xiangyu Jia, Zhibo Chen, Shengyi Zhang, and Xiaojing Xue

***Abstract*—Remote sensing change detection (RSCD) is widely used for monitoring land-surface dynamics. Most models, however, output an unconditional "all-changes" map and do not account for user intents expressed in natural language. This can lead to spurious masks under counterfactual (i.e., negative/mismatched) queries. This article studies language-guided reasoning change detection. Given a bitemporal image pair and a text instruction, the model outputs an instruction-conditioned change mask; the correct output can be an empty mask when the queried change is absent. To support this setting, ReasonRS-110K is built as a benchmark of 112 317 image–instruction–mask triplets. Instructions are organized into hierarchical reasoning levels (L1–L3) together with L0 negative queries for compliance evaluation. Change-LISA is proposed as a parameter-efficient framework that combines a frozen Siamese SAM encoder and an explicit difference/fusion module with a multimodal LLM (LLaVA). The LLM emits a `<SEG>`token whose embedding prompts a lightweight mask decoder. Only low-rank adaptation (LoRA) adapters and the decoder are optimized. On ReasonRS-110K, Change-LISA reaches 63.1 IoU and 77.3 $F1$ on positive-query levels (L1–L3) and 95.4 NCA with 0.12 FPA on L0 negative queries, outperforming representative change detection (CD)-only, referring image segmentation (RIS), and multimodal large language models (MLLMs) baselines. The code for this work is available at https://github.com/jxy0413/changeLisa**

***Index Terms*—Counterfactual, instruction-conditioned segmentation, low-rank adaptation (LoRA), multimodal large language model, negative query, parameter-efficient fine-tuning, remote sensing change detection (RSCD), segment anything model (SAM), vision-language model (VLM).**

I. Introduction

REMOTE sensing change detection (RSCD) plays a vital role in monitoring land-surface dynamics from multitemporal observations. It supports applications such as urban expansion analysis [1], [2], environmental surveillance [3], [4], and rapid postdisaster assessment [5]. With the increasing availability of high-resolution satellite and aerial imagery, modern RSCD models have achieved substantial progress in pixel-level change localization [6], [7], [8]. In practice, decision-making often requires a selective understanding of changes tied to a specific user intent or semantic goal (e.g., damaged buildings after a wildfire or flooded residential areas after a storm [9]).

Most existing RSCD methods are trained to produce an unconditional change map, i.e., segmenting all apparent differences between two timestamps without considering the user's intent [10]. Many differences (e.g., seasonal variation, illumination/shadow changes, and sensor-induced appearance shifts) are irrelevant to a specific analysis goal and may lead to distracting false alarms [11]. Users often ask semantic, intent-driven questions—such as "buildings destroyed by wildfire" or "flood-impacted residential areas"—that require selecting only the queried changes and ignoring others [12]. This mismatch motivates instruction-conditioned change detection that follows natural-language queries rather than blindly segmenting all changes.

A straightforward direction is to borrow language grounding techniques from referring image segmentation (RIS) and vision-language models (VLMs) [13], which align natural-language expressions with image regions [14]. Most RIS methods are designed for single-image grounding and lack explicit temporal discrepancy modeling, making it difficult to separate "what changed" from "what is salient" when applied to bitemporal imagery. Recent multimodal LLMs for remote sensing provide instruction understanding and coarse grounding. Their outputs are typically text answers or sparse localization cues, so extra pipelines are needed to obtain pixel-level masks [15]. These models are also rarely evaluated under counterfactual/negative queries where the correct output should be an empty mask, so instruction compliance and hallucination-like false positives remain underexamined [16]. Relatedly, recent CDQAG frameworks jointly produce textual answers and pixel-level grounding masks across diverse change questions [17], [18]; by contrast, we study instruction-conditioned change segmentation with an explicit counterfactual (L0) compliance protocol and dedicated metrics (NCA/FPA).

A. What We Mean by Counterfactual Queries

In this work, "counterfactual queries" refer to L0 negative (mismatched) instructions whose semantics are *not* satisfied by the given bitemporal pair, i.e., no change region in (x_{T1}, x_{T2}) matches the queried concept. Formally, let $S(x_{T1}, x_{T2}, q)$ denote the set of pixels that: 1) changed between $T1$ and $T2$ and 2) satisfy the instruction q; a counterfactual query corresponds to $S = \varnothing$, and the ground-truth (GT) mask is the all-zero mask $M = \mathbf{0}$. We construct such L0 queries by

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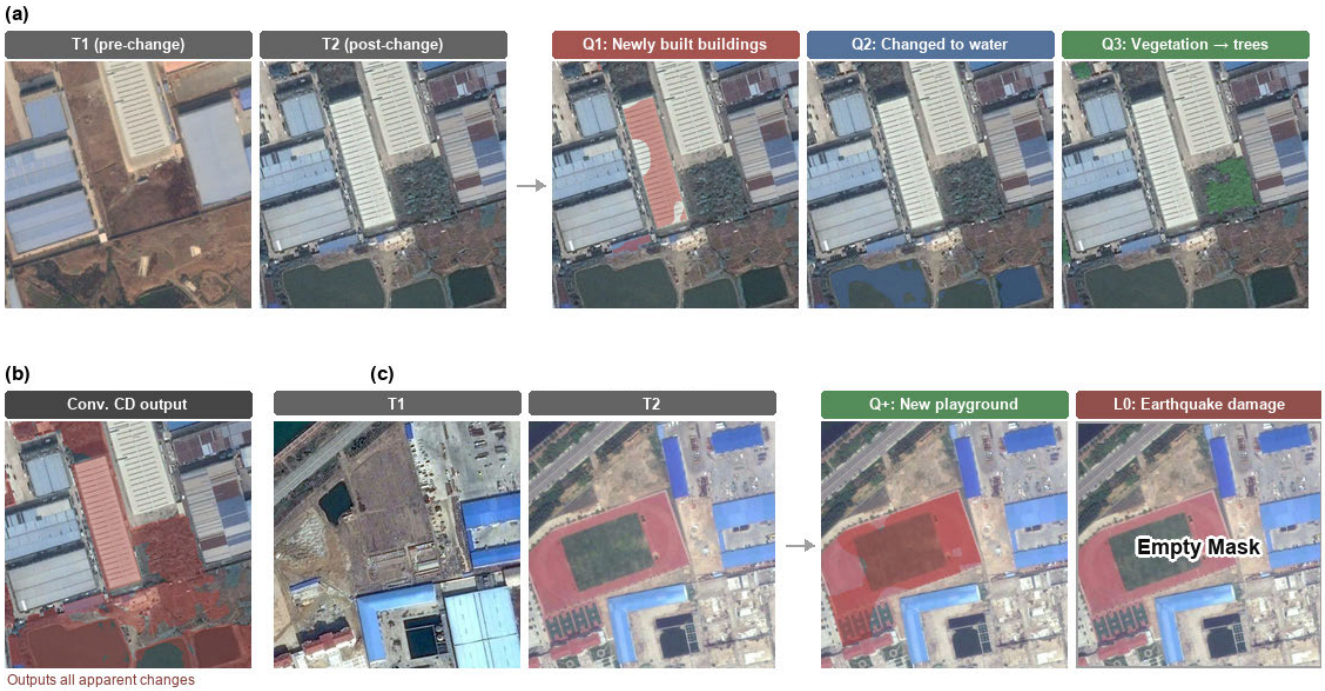


Fig. 1. Illustration of the proposed reasoning semantic change detection task. Given bitemporal images and a text instruction, the model must produce a query-conditioned change mask (or an empty mask for negative/counterfactual queries). (a) For the same image pair, different queries lead to different selective change masks (e.g., newly built buildings, changed to water, and vegetation→trees). (b) Conventional change detection outputs all apparent changes regardless of the query. (c) Our setting includes both positive queries (producing a nonempty mask) and L0 negative/counterfactual queries where the correct output is an *empty* mask.

pairing image pairs with semantically mismatched instructions (Section III-G), including them during training to enforce abstention, and evaluating compliance using NCA/FPA (Section IV-B).

B. Why Naive Early Fusion Is Insufficient

One may consider a straightforward adaptation by stacking or concatenating x_{T1} and x_{T2} and feeding them into a single-image RIS model (e.g., LISA) together with the instruction. This early-fusion strategy often suffers from *temporal ambiguity*. Without an explicit discrepancy representation, the model can respond to visually salient objects that exist in either timestamp rather than the *changed* regions, and it may confuse "appeared" versus "disappeared" targets under language constraints. This issue is exacerbated in remote sensing due to misregistration, illumination/seasonal shifts, and spectral variability (same-object different-spectrum/different-object same-spectrum). In these cases, raw appearance differences can dominate and induce spurious masks, especially for counterfactual queries that should yield an empty output. These observations motivate an explicit temporal interaction and difference modeling component that separates *change cues* from *semantic selection*.

To bridge this gap, we formulate language-guided reasoning change detection in remote sensing [19], [20]. Given a bitemporal image pair and a natural-language instruction, the model is required to output a query-conditioned change mask that highlights only the changes consistent with the instruction. Importantly, the correct prediction can be an empty mask when the instruction is counterfactual, or no relevant change exists in the scene, which directly tests instruction compliance and reduces hallucination-like spurious outputs [21], [22]. As illustrated in Fig. 1, this task requires both fine-grained localization and semantic/causal reasoning under language queries, including the ability to output an empty mask for negative instructions. To characterize diverse cognitive demands, we organize queries into hierarchical reasoning levels, ranging from basic semantic transitions to higher-level attribute/context understanding and causal/functional reasoning [13], [23]. Accordingly, we evaluate models in a level-wise manner, reporting conventional segmentation quality on positive queries (L1–L3) and compliance-oriented metrics on negative queries (L0) [24].

To support this task [1], we construct ReasonRS-110K, a large-scale benchmark that integrates multiple public change detection datasets to cover diverse scenes and change types. Mask-conditioned descriptions are generated with a VLM and rewritten into diverse natural-language instructions spanning hierarchical reasoning levels. Expert verification and filtering are applied to improve reliability. L0 counterfactual queries are constructed by pairing images with mismatched instructions. The resulting dataset contains over 112k image–instruction–mask triplets with an explicit level-wise taxonomy, enabling systematic training and evaluation of instruction-following change segmentation. Building on this benchmark, we propose Change-LISA, a parameter-efficient framework that adopts a frozen Siamese SAM encoder for dense spatial priors, an explicit difference/fusion module to capture temporal discrepancies, and a multimodal LLM

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(LLaVA) to emit a <SEG> token as a semantic prompt. Conditioned on the <SEG> embedding, a lightweight mask decoder produces pixel-accurate change masks, while only low-rank adaptation (LoRA) adapters and the decoder are optimized to ensure efficient adaptation and instruction compliance.

In summary, our contributions are threefold.

1) We formulate language-guided reasoning change detection for remote sensing and introduce a hierarchical query taxonomy with counterfactual/negative queries to evaluate instruction compliance.
2) We construct ReasonRS-110K (112 317 ~ 110 K) as a large-scale benchmark with pixel-level supervision, level-wise evaluation (L1–L3), and compliance-oriented metrics on L0.
3) We propose Change-LISA, a parameter-efficient framework that integrates a Siamese SAM encoder, explicit temporal difference fusion, and an LLM-prompted pixel-level decoder for instruction-conditioned change segmentation, and validate its effectiveness against change detection (CD)-only, RIS, and multimodal large language models (MLLMs) baselines.

C. Data and Reproducibility

To facilitate community research and independent auditing, we will release ReasonRS-110K together with the instruction generation/rewriting rules and evaluation code upon acceptance/publication.

II. Related Work

A. Remote Sensing Change Detection

RSCD aims to localize and segment meaningful land-surface changes from bitemporal remote sensing imagery [25]. It supports applications such as urban expansion monitoring [26] and postdisaster assessment [27]. Early RSCD methods mainly adopted Siamese CNN architectures with feature differencing to model cross-temporal changes [28] (e.g., FC-Siam-Diff [29]). Recent advances introduced attention/Transformer-based architectures (BIT [7] and ChangeFormer [24]) and started to leverage foundation segmentation models like SAM [30] as spatial priors for change representation. Beyond fully supervised settings, recent studies have explored label-efficient RSCD to alleviate the high cost of dense pixel annotations. For example, SWCD adopts similarity-awareness weakly supervised learning to improve change localization from limited supervision, while SCAD-Net develops a semi-supervised connectivity-aware decoupling strategy to exploit unlabeled data and preserve structural consistency in high-resolution scenes [31], [32]. Meanwhile, multimodal change detection has also attracted increasing attention. ChangeCLIP introduces vision-language representation learning into RSCD to enhance semantic change modeling, while MMChange and related multimodal feature fusion methods further incorporate textual difference cues into the change detection pipeline [33], [34]. These approaches demonstrate the value of weak supervision and multimodal semantics for conventional RSCD; however, they still mainly target unconditional change mapping rather than instruction-conditioned selective segmentation. Most existing RSCD models are purely vision-driven and lack language-conditioned instruction following, making it difficult to selectively segment only the changes that match complex semantic or causal queries [35], [36]. They also perform poorly under counterfactual or negative queries where no region should be returned. Text-agnostic models often produce nonempty change masks, resulting in false positives and poor compliance [7]. This limitation motivates instruction-conditioned reasoning change detection that can localize relevant changes and output an empty mask when the queried change is absent [37].

B. Referring Image Segmentation

RIS aims to segment the pixel-level region in an image that corresponds to a given natural-language expression [38]. Recent RIS methods typically build strong vision-language alignment via cross-modal fusion and attention mechanisms, achieving accurate language-grounded segmentation in natural images (e.g., LAVT [39] and GRES [40]). More recently, LISA [41] bridges multimodal LLM reasoning and dense prediction by introducing an embedding-as-mask paradigm, where the model generates a special <SEG> token whose hidden embedding prompts a lightweight decoder to produce a pixel-accurate mask. In remote sensing, language-guided segmentation has gained traction, with works such as RemoteSAM [42] and RRSIS [37], along with prompt-based SAM adaptations (e.g., RSPrompter) and grounded/foundation VLMs (e.g., GeoChat [15], SkySense [43]). Most of these approaches are designed for single-image grounding and lack explicit temporal discrepancy modeling. When adapted to bitemporal change detection, they may blur temporal ordering and produce spurious or boundary-bleeding masks, especially under counterfactual/negative queries that should output an empty mask [44].

C. Multimodal LLMs for Remote Sensing

MLLMs for remote sensing aim to unify vision and language to support open-vocabulary scene understanding [45], question answering, and reasoning over aerial imagery. Recent works (e.g., GeoChat [15], RSGPT [46], EarthGPT [47], and SkyEyeGpt [48]) show promising instruction understanding and coarse grounding in geospatial scenarios. Their outputs are often textual responses or coarse localization cues (e.g., boxes/points) rather than pixel-accurate masks, so a grounding-to-mask pipeline (e.g., prompting SAM with predicted boxes) is typically required. This extra step can accumulate errors and may miss fine boundaries in complex remote sensing scenes. Most MLLMs are also not designed for bitemporal change understanding and thus lack structured temporal discrepancy modeling under instruction constraints, which can hurt reliability for counterfactual/negative queries where the correct output is an empty mask [16]. Recent CDQAG works further explore joint text answering and pixel-level change grounding [17], [18]; our setting is complementary and emphasizes instruction-conditioned change mask prediction,

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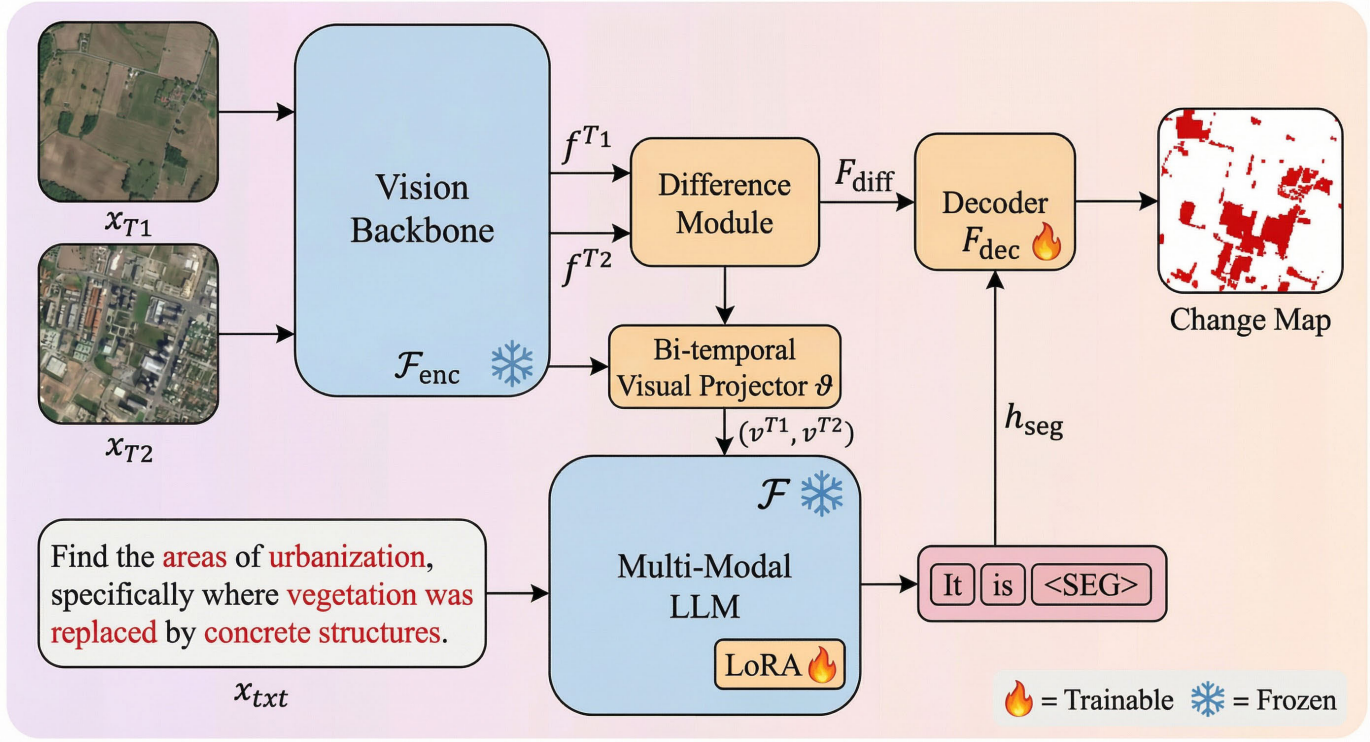


Fig. 2. The pipeline of Change-LISA. We adopt a Siamese vision backbone ($\mathcal{F}_{enc}$) to extract visual features from bitemporal images. These features interact with the text instruction in the multimodal LLM ($\mathcal{F}$) to generate the segmentation token <SEG>. Simultaneously, a difference/fusion module ($\mathcal{F}_{diff}$) performs temporal fusion by computing feature disparities between x_{T1} and x_{T2}. The decoder ($\mathcal{F}_{dec}$) then takes the embedding of <SEG> (h_{seg}) as a prompt to decode the change map $\hat{M}$ from the difference features. Note that we only tune the LoRA layers and the decoder (indicated by fire), while keeping most parameters frozen (indicated by snowflake).

explicit temporal discrepancy modeling, and metric-driven counterfactual compliance evaluation (L0 with NCA/FPA).

D. Datasets and Benchmarks

RSCD benchmarks typically provide bitemporal image pairs with pixel-level change masks to support training and evaluation under the conventional BCD/SCD setting (e.g., SECOND [49], LEVIR-CD [10], xBD [50], and Hi-UCD [51]). These datasets have driven substantial progress in change localization, but they are generally formulated to produce unconditional change maps. They also do not include natural-language queries that specify what changes should be selected. In parallel, recent remote sensing vision-language datasets and models (e.g., Qwen3-VL [52]) have improved image–text understanding and conversational analysis, but they often lack dense supervision or task protocols tailored to change segmentation. Existing benchmarks also rarely incorporate counterfactual/negative queries where the correct output is an empty mask, and thus provide limited means to quantify instruction compliance and hallucination-like false positives. To address these gaps, we introduce ReasonRS-110K, a large-scale benchmark for language-guided reasoning change detection with a hierarchical query taxonomy (L1–L3) and explicit L0 negative queries. ReasonRS-110K further couples pixel-level supervision with compliance-oriented metrics (e.g., NCA/FPA), enabling systematic evaluation of both segmentation quality and the ability to abstain when the queried change is absent.

III. Methodology

This section presents Change-LISA for language-guided reasoning change detection in remote sensing. Given a bitemporal image pair (x_{T1}, x_{T2}) and a natural-language query q, the goal is to predict a query-conditioned change mask M, which can be empty for negative or counterfactual queries. As shown in Fig. 2, the framework uses a Siamese vision encoder with shared weights and a bi-temporal interaction and difference/fusion module $\mathcal{F}_{diff}$ (Section III-D) to inject change cues before language-conditioned mask decoding. The model is trained with segmentation supervision and query-following constraints, and evaluated on ReasonRS-110K with a hierarchical query taxonomy and compliance-oriented metrics. The rest of this section is organized as follows. Section III-A defines the task; Sections III-B–III-F describe the model and training; and Section III-G details the dataset and protocol.

A. Task

We formulate the task of language-guided reasoning change detection in remote sensing. Given a bitemporal image pair $\mathcal{X}_{img} = (x_{T1}, x_{T2})$ and a natural-language instruction x_{txt}, the goal is to predict a binary change mask $\hat{M} \in \{0,1\}^{H\times W}$ that satisfies the instruction

$$\hat{M} = \mathcal{F}_\theta(x_{T1}, x_{T2}, x_{txt}) \quad (1)$$

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where $\mathcal{F}_\theta$ represents the reasoning model parameterized by θ. Unlike conventional change detection (BCD/SCD) that blindly segments all appearing differences, our task requires the model to comprehend implicit user intentions and segment *only* the changes consistent with x_{txt}. We also introduce negative (counterfactual) queries, where no region in the images satisfies the instruction; in such cases, the GT is an *empty* mask, implemented as an all-zero mask ($M = \mathbf{0} \in \{0,1\}^{H\times W}$). To evaluate robustness, queries are categorized into hierarchical reasoning levels, ranging from basic semantic transitions (e.g., vegetation to concrete) to high-level causal/functional reasoning (e.g., disaster impact), with negatives treated as a distinctive capability check.

B. Overview

Fig. 2 illustrates the overall pipeline of Change-LISA. Given a bitemporal image pair $\mathcal{X}_{img} = (x_{T1}, x_{T2})$ and a natural-language instruction x_{txt}, our goal is to predict a query-conditioned change mask $\hat{M}$. Specifically, we first adopt a frozen Siamese vision backbone $\mathcal{F}_{enc}$ with shared weights to extract dense visual representations from the two timestamps. These visual features are then bifurcated into two paths: one feeds into a difference module $\mathcal{F}_{diff}$ to compute temporal discrepancies and preserve fine-grained change details; the other is projected into the multimodal LLM $\mathcal{F}$ to align with the text instruction. The LLM subsequently performs reasoning and generates a specific segmentation token <SEG>, whose hidden embedding is denoted as h_{seg}. Finally, conditioned on h_{seg}, the decoder $\mathcal{F}_{dec}$ fuses the semantic guidance with the difference features to generate the final change map $\hat{M}$. A parameter-efficient training scheme is used by exclusively tuning the LoRA layers and the decoder, enabling instruction following while maintaining generalization capabilities.

C. Vision Encoder

We adopt the segment anything model (SAM [30]) with a ViT-H backbone as our Siamese vision encoder $\mathcal{F}_{enc}$. The encoder is shared across the two timestamps and kept frozen to preserve general visual representations and ensure stable cross-temporal alignment [53]. Given a bitemporal image pair $\mathcal{X}_{img} = (x_{T1}, x_{T2})$, we extract dense visual features as follows:

$$f_{T1} = \mathcal{F}_{enc}(x_{T1}), \quad f_{T2} = \mathcal{F}_{enc}(x_{T2}) \tag{2}$$

where f_{T1} and f_{T2} are dense feature maps capturing spatial structures and fine-grained details. Using shared weights constrains both timestamps to the same feature space, which benefits temporal comparison while avoiding unnecessary parameter growth. The resulting dense features serve as the interface to subsequent modules. On the change modeling path, (f_{T1}, f_{T2}) are fed into the difference module $\mathcal{F}_{diff}$ to form change-aware representations that preserve local details. In parallel, the visual features are projected and provided to the multimodal LLM $\mathcal{F}$ together with the instruction x_{txt}, enabling language-conditioned reasoning and the generation of the segmentation prompt embedding for mask decoding. This design separates dense spatial encoding from language-driven selection, providing accurate localization cues for instruction-following change segmentation.

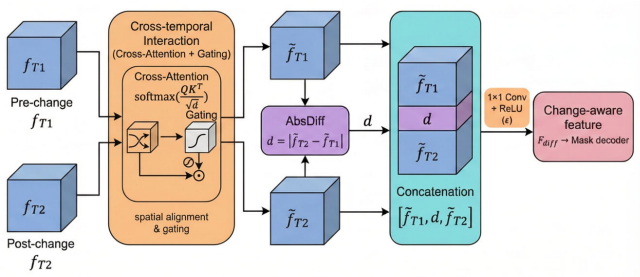


Fig. 3. Bitemporal interaction and difference/fusion module. The module first aligns bitemporal features via cross-temporal cross-attention and sigmoid gating fusion, then computes an explicit absolute difference and fuses aligned features with the difference to produce a change-aware representation F_{diff} for mask decoding.

TABLE I
LEVEL-WISE STATISTICS OF REASONRS-110K

Level	#Samples	Ratio (%)
L0 Negative	13,882	12.36
L1 Semantic Transition	34,130	30.40
L2 Attribute Context	51,982	46.28
L3 Causal Functional	12,323	10.97
Total	**112,317**	**100.00**

D. Bi-Temporal Interaction and Difference/Fusion Module

To explicitly model temporal discrepancies and suppress spurious appearance variations, we design a lightweight bitemporal interaction and difference/fusion module $\mathcal{F}_{diff}$ (Fig. 3). Given dense features $f_{T1}, f_{T2} \in \mathbb{R}^{H\times W\times C}$, we first apply a cross-temporal interaction (cross-attention + gating) to obtain aligned features. We flatten the spatial dimensions into $N = HW$ tokens

$$F_{T1} = \text{reshape}(f_{T1}) \in \mathbb{R}^{N\times C}$$
$$F_{T2} = \text{reshape}(f_{T2}) \in \mathbb{R}^{N\times C}. \tag{3}$$

1) Mechanism for Suppressing Spurious Variations: Note that (3) only performs tokenization (reshape) and does not suppress spurious variations by itself. The suppression is realized by the subsequent *alignment-before-differencing* process: cross-temporal attention retrieves temporally corresponding features to mitigate pseudo-differences caused by slight misregistration/viewpoint shifts, while the MLP-based sigmoid gate adaptively filters components dominated by global appearance changes (e.g., illumination/seasonal/sensor shifts) when fusing F and Z. Therefore, when the absolute difference is computed on the aligned features, nuisance-induced discrepancies are already attenuated, leading to fewer spurious responses (as supported by our ablation studies). We then compute cross-temporal cross-attention (shown for $T1 \leftarrow T2$) to retrieve temporally corresponding features

$$Q_1 = F_{T1}W_Q, \quad K_2 = F_{T2}W_K, \quad V_2 = F_{T2}W_V$$
$$A_{1\leftarrow 2} = \text{softmax}\left(\frac{Q_1 K_2^\top}{\sqrt{D}}\right), \quad Z_{T1} = A_{1\leftarrow 2}V_2 \tag{4}$$

where $W_Q, W_K, W_V \in \mathbb{R}^{C\times D}$ are learnable projections and softmax$(\cdot)$ is applied along the key/token dimension. Finally,

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TABLE II
HIERARCHICAL QUERY TAXONOMY AND OPERATIONAL CRITERIA ON REASONRS-110K

Level	Definition	Operational criteria & visual evidence	Instruction examples
L0 (Negative)	**Compliance & Abstention.** Output an empty mask when no visual evidence matches the query.	**Rule:** Query semantics are disjoint from the ground-truth change. **Constraint:** Distractors may exist but must be ignored. **Cues:** Mismatch between query and scene.	• "Segment areas **flooded** between T1 and T2." *Scene: urban construction, no water* • "Find buildings **destroyed** by wildfire." *Scene: healthy vegetation, no burn scars*
L1 (Semantic)	**Category Transition.** Detect change based on land-cover class shifts (A→B).	**Rule:** Change is defined solely by category. **Constraint:** No fine-grained attributes or event inference. **Cues:** Distinct texture/color shift (vegetation→concrete).	• "Find newly constructed **buildings**." • "Segment regions changed from **forest** to **bare land**."
L2 (Attribute)	**Attribute & Context.** Reason about subtypes, attributes, or spatial context.	**Rule:** Targets specific attributes (industrial vs. residential) or spatial context. **Constraint:** Requires filtering by description. **Cues:** Roof patterns, greenhouse stripes, road geometry.	• "Segment newly built **industrial** buildings." • "Locate newly added **greenhouse** facilities."
L3 (Causal)	**Event & Function.** Infer cause (event) or functional consequence of change.	**Rule:** Explicitly references cause (event) or impact. **Constraint:** Requires linking visual traces to abstract events. **Cues:** Flood patterns, burn scars, debris footprints.	• "Segment areas **affected** by **flooding**." • "Find buildings **likely destroyed** by an **earthquake**."

Note: L0 queries are counterfactual with empty masks; L1–L3 are positive queries with increasing reasoning demands.

TABLE III
INSTRUCTION GENERATION, FILTERING, AND VERIFICATION PIPELINE FOR REASONRS-110K.WE USE A VLM TO PRODUCE MASK-CONDITIONED DESCRIPTIONS AND REWRITE THEM INTO LEVEL-WISE INSTRUCTIONS, FOLLOWED BY FILTERING AND MANUAL REVIEW FOR QUALITY CONTROL

Stage	Input / Output	Key design / rules
Step 1: Description generation	$(x_{T1}, x_{T2}, M) \rightarrow$ change description	**VLM:** Qwen3-VL (`qwen3-vl-plus`, T=0.5). **Input:** concatenated visual input $[x_{T1} \| x_{T2}]$ with red overlay highlighting M on $T2$. **Goal:** conservative description of *what changed*.
Step 2: Level-wise rewriting	Description → instruction set	**L1:** direct category phrasing ("X changed to Y"). **L2:** inject attribute keywords (e.g., "industrial", "near river"). **L3:** enforce causal wording ("due to", "caused by") via few-shot examples; allow "likely" only when visually supported.
Step 3: Automatic filtering	Raw → cleaned instructions	**Length:** 5–30 tokens. **Hallucination:** ban invisible traits ("ownership", "age"). **Ambiguity:** reject multi-target or vague referents. **Retention:** $\sim$85% after filtering.
Step 4: Human verification	Stratified 5% → refined rules	**Annotators:** 2 RS-domain reviewers + 1 arbitrator. **Checklist:** mask match, verifiability, clarity. **Top errors:** hallucinated attribute, wrong direction, ambiguous referent. **Pass rate:** $\sim$89.8%; failures trigger prompt/filter refinement.
Step 5: L0 negative construction	Image pair + mismatched query → empty mask	**Cross-dataset pairing:** disaster queries paired with non-disaster scenes. **Keyword exclusion:** same semantic group forbidden (e.g., "flood" excluded from water-body scenes). **Screening:** 5% spot-check; $\sim$8% ambiguous pairs discarded.

TABLE IV
SOURCE DISTRIBUTION OF REASONRS-110K

Source	Ratio (%)	Primary contribution
SECOND	63.9	Semantic variety (multi-class land-cover)
xBD	13.7	Disaster scenarios (building damage)
LEVIR-CD	11.0	Urban construction changes
Hi-UCD	11.4	Specialized urban attributes

a sigmoid gate adaptively fuses the original tokens with the attended tokens

$$G_{T1} = \sigma\left(\mathrm{MLP}\left([F_{T1}, Z_{T1}, F_{T1} - Z_{T1}]\right)\right)$$
$$\tilde{F}_{T1} = G_{T1} \odot F_{T1} + (1 - G_{T1}) \odot Z_{T1}. \quad (5)$$

We symmetrically compute Z_{T2} and $\tilde{F}_{T2}$ via $T2 \leftarrow T1$, and reshape back to obtain $\tilde{f}_{T1}, \tilde{f}_{T2} \in \mathbb{R}^{H \times W \times C}$. We then compute an explicit difference map via absolute feature difference

$$d = \left|\tilde{f}_{T2} - \tilde{f}_{T1}\right|. \quad (6)$$

Finally, we fuse the aligned features and the explicit difference through concatenation and a 1×1 convolution (with ReLU) to form a change-aware representation

$$F_{diff} = \phi\left(\left[\tilde{f}_{T1}, \tilde{f}_{T2}, d\right]\right) \quad (7)$$

where $[\cdot]$ denotes channel-wise concatenation and $\phi(\cdot)$ is implemented as a 1×1 Conv + ReLU. The resulting F_{diff} preserves fine-grained boundaries while emphasizing temporally inconsistent regions, and is subsequently fed to the mask

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TABLE V

MAIN RESULTS ON REASONRS-110K. WE REPORT IOU AND $F1$ SCORES FOR POSITIVE QUERY LEVELS (L1–L3) AND NCA/FPA FOR NEGATIVE QUERIES (L0). "OVERALL" DENOTES THE SAMPLE-WEIGHTED AVERAGE. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Overall (L1–L3)		L1 (Semantic)		L2 (Attribute)		L3 (Causal)		L0 (Negative)	
	IoU	F1	IoU	F1	IoU	F1	IoU	F1	NCA ↑	FPA ↓
Generic Change Detection (CD-only)										
FC-Siam-Diff [29]	28.5	44.3	40.2	57.1	25.4	40.5	19.8	33.0	0.00	9.12
SNUNet [55]	35.8	52.7	51.5	67.9	32.6	49.1	23.4	37.9	0.00	8.45
BIT [7]	38.2	55.3	54.8	70.8	34.2	50.9	25.6	40.7	0.00	7.20
ChangeFormer [24]	41.5	58.7	59.3	74.4	38.5	55.6	28.4	44.2	0.00	6.55
Referring Image Segmentation (RIS)										
LAVT [39]	42.6	59.7	50.4	66.8	42.1	59.2	35.3	52.1	45.1	4.26
LISA (orig.) [41]	45.2	62.1	52.1	68.3	43.6	60.5	38.5	55.4	48.5	3.20
GRES [40]	46.5	63.2	53.2	69.5	45.8	62.4	38.4	55.1	72.5	1.50
RemoteSAM [42]	50.1	66.5	55.2	70.9	50.1	66.7	45.6	62.5	52.1	2.80
Multimodal Large Models (MLLMs)										
LLaVA-v1.5 [54] + SAM [30]	30.5	46.2	35.1	51.5	30.5	46.2	25.3	40.1	45.0	3.50
GeoChat [15] + SAM [30]	37.8	54.3	42.5	59.3	38.2	54.8	33.1	49.2	60.5	2.10
Change-LISA (Ours)	**63.1**	**77.3**	**68.5**	**81.2**	**62.1**	**76.5**	**58.7**	**73.2**	**95.4**	**0.12**

TABLE VI

ZERO-SHOT RESULTS ON SYSU-CD (UNSEEN DURING TRAINING). WE EVALUATE ON THE FULL SYSU-CD TEST SPLIT (4000 PAIRS), REPORTING IOU/$F1$ ON POSITIVE QUERIES (L1–L3) AND NCA/FPA ON NEGATIVE QUERIES (L0)

Method	L1–L3 (Positive)		L0 (Negative)	
	IoU	F1	NCA ↑	FPA ↓
ChangeFormer [24]	19.8	32.4	0.00	16.20
RemoteSAM [42]	31.5	47.2	18.5	8.15
GeoChat [15] + SAM [30]	24.6	38.5	28.4	6.50
Change-LISA (Ours)	**43.2**	**59.8**	**75.6**	**1.45**

TABLE VII

ABLATION STUDY ON REASONRS-110K. WE INVESTIGATE THE IMPACT OF THE EXPLICIT DIFFERENCE FUSION, LLM ADAPTATION (LORA), AND NEGATIVE SAMPLE TRAINING. "FULL" DENOTES OUR PROPOSED CHANGE-LISA

Variant	Diff/Fusion	LLM Adapt.	L0 Train	IoU (L1–L3)	F1 (L1–L3)	NCA (L0)↑	FPA (L0)↓
w/o Diff/Fusion	No	LoRA	Yes	60.5	74.8	93.2	0.45
Frozen LLM (w/o LoRA)	Yes	None	Yes	57.2	71.5	89.8	0.92
w/o L0 Training	Yes	LoRA	No	62.8	76.9	0.00	12.50
Change-LISA (Full)	Yes	LoRA	Yes	**63.1**	**77.3**	**95.4**	**0.12**

decoder $\mathcal{F}_{dec}$ (together with the semantic prompt h_{seg}) for query-conditioned change mask prediction.

2) Design Rationale: We choose *AbsDiff* + *Concat* for two complementary reasons. First, $d = |\tilde{f}_{T2} - \tilde{f}_{T1}|$ acts as an explicit high-frequency discrepancy cue that highlights boundary-aligned changes and reduces the burden on the LLM/decoder to "discover" differences implicitly. Second, concatenating $(\tilde{f}_{T1}, \tilde{f}_{T2})$ retains appearance context (what the region looks like before/after), which is crucial for language-conditioned selection (e.g., "newly built greenhouses" rather than any generic change). The preceding alignment/gating further mitigates RS-specific nuisances such as small misregistration and viewpoint-induced shifts by encouraging correspondence before differencing; as a result, the module suppresses pseudo-changes while maintaining sharp edges, improving both positive-query masks and empty-mask compliance under counterfactual queries.

E. Language Adapter

Most multimodal LLMs possess strong capabilities in understanding images and text instructions, yet they lack the ability to directly output fine-grained segmentation masks. To bridge this gap, following the "embedding-as-mask" paradigm

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TABLE VIII
STRUCTURAL ABLATION OF $\mathcal{F}_{diff}$. ALL RESULTS ARE ON REASONRS-110K UNDER THE SAME SETTINGS AS SECTION IV-C

Variant	L1–L3		L0	
	IoU	F1	NCA↑	FPA↓
AbsDiff	58.8	73.5	94.5	0.25
Concat	61.2	75.6	93.5	0.42
Concat+AbsDiff	62.4	76.8	95.0	0.18
Ours	**63.1**	**77.3**	**95.4**	**0.12**

in LISA, we introduce a language adapter that translates high-level reasoning into a semantic embedding for mask decoding.

1) Multimodal LLM and Parameter-Efficient Tuning: We employ LLaVA-v1.5 [54] as the core multimodal LLM $\mathcal{F}$. To ensure efficient adaptation to the remote sensing domain while preventing catastrophic forgetting, we freeze the vast majority of LLM parameters and exclusively optimize lightweight LoRA modules injected into the attention layers, as depicted in Fig. 2.

2) Bitemporal Visual Injection: Given the dense visual features f_{T1} and f_{T2} extracted by the Siamese encoder, we first align them into the LLM's word embedding space via a learnable linear projector $\gamma(\cdot)$

$$v_{T1} = \gamma(f_{T1}), \quad v_{T2} = \gamma(f_{T2}). \tag{8}$$

We concatenate these bitemporal tokens with the text instruction embeddings x_{txt} to form the input sequence. This allows the LLM to perform implicit cross-temporal interaction via its self-attention mechanism, effectively comparing the semantic content of $T1$ and $T2$ under the guidance of the instruction.

3) <SEG> Token as a Reasoning Anchor: During the autoregressive generation, when the model decides to segment the target change, it generates a special token <SEG>. We extract the last-layer hidden embedding of this token to serve as the global context prompt, denoted as h_{seg}

$$h_{seg} = \mathcal{F}(x_{txt}, v_{T1}, v_{T2})\big|_{<\text{SEG}>}. \tag{9}$$

4) Prompted Mask Decoding: Finally, the embedding h_{seg} acts as a dynamic condition for the mask decoder. It guides the decoder to pinpoint the change regions from the high-resolution difference features

$$\hat{M} = \mathcal{F}_{dec}\left(h_{seg}, \mathcal{F}_{diff}(f_{T1}, f_{T2})\right). \tag{10}$$

Through this design, the LLM provides semantic reasoning (What to segment?), while the difference module and decoder ensure spatial precision (Where is the boundary?).

F. Training

We train the proposed Change-LISA in an end-to-end manner employing a parameter-efficient tuning strategy. To preserve the generalization capability of pretrained foundations, we freeze the Siamese vision backbone $\mathcal{F}_{enc}$ (SAM ViT-H) and the multimodal LLM $\mathcal{F}$ (LLaVA-v1.5). Only the injected LoRA modules, the difference module $\mathcal{F}_{diff}$, and the mask decoder $\mathcal{F}_{dec}$ are optimized, as illustrated in Fig. 2.

1) Training Data Formulation: Each training sample is organized as a quadruple $\langle x_{T1}, x_{T2}, x_{txt}, M\rangle$, consisting of the bitemporal image pair, the natural-language instruction, and the GT binary change mask. We unify the supervision format using a conversation template, where the assistant's response is required to contain the special token <SEG> to trigger segmentation. Our training set includes both positive queries ($M \neq \emptyset$) and negative (counterfactual) queries ($M = \emptyset$). This mixture encourages the model to follow instructions and suppress false positives when no relevant change exists.

2) Optimization Objective: Following the embedding-as-mask paradigm, the overall objective $\mathcal{L}$ is a weighted sum of the text generation loss and the mask prediction loss:

$$\mathcal{L} = \lambda_{txt}\mathcal{L}_{txt} + \lambda_{mask}\mathcal{L}_{mask}. \tag{11}$$

The text loss $\mathcal{L}_{txt}$ is the standard auto-regressive cross-entropy loss, supervising the LLM to generate the correct response and the <SEG> token. The mask loss $\mathcal{L}_{mask}$ evaluates the quality of the predicted mask $\hat{M}$. We employ a combination of per-pixel binary cross-entropy (BCE) loss and Dice loss

$$\mathcal{L}_{mask} = \lambda_{bce}\text{BCE}\left(\hat{M}, M\right) + \lambda_{dice}\text{Dice}\left(\hat{M}, M\right). \tag{12}$$

For negative queries where the GT M is an all-zero mask, this loss objective explicitly penalizes any nonzero predictions, effectively teaching the model to output "nothing" when the instruction does not match the visual changes.

G. Data and Protocol

We build the benchmark by integrating four diverse public datasets: SECOND [49], LEVIR [10], xBD [50], and Hi-UCD [51]. Given the bitemporal pairs and masks, we utilize a VLM (Qwen3-VL [52]) to generate mask-conditioned descriptions, which are then rewritten into diverse instructions covering L1–L3 via prompt engineering. To improve reliability, we further apply expert verification and filtering: we randomly sample 5% of the generated data for manual review to identify hallucinated attributes and ambiguous instructions, and refine the prompts/filters accordingly. To test instruction compliance, we construct L0 negative queries by pairing images with mismatched instructions.

The resulting ReasonRS-110K benchmark comprises a total of 112 317 (~110 K) image-instruction-mask triplets. To ensure a comprehensive evaluation across different cognitive demands, the data is distributed as follows (see Fig. 4): 34 130 samples for L1 Semantic Transition, providing a solid foundation for land-cover changes; 51 982 samples for L2 Attribute Context, which constitutes the largest portion to cover diverse urban attributes (see Table I); and 12 323 samples for high-level L3 Causal Functional reasoning, focusing on complex disaster and infrastructure scenarios. Additionally, 13 882 L0 Negative samples are incorporated to robustify the model against hallucinations. In terms of data sources (as shown in Table IV), SECOND contributes the majority of semantic variety (63.9%), complemented by xBD (13.7%), LEVIR-CD (11.0%), and Hi-UCD (11.4%) for specialized reasoning tasks. We follow a random split of approximately 8:1:1 for training, validation, and testing.

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IV. Experiments

A. Datasets

We evaluate Change-LISA on the proposed ReasonRS-110K benchmark for language-guided reasoning change detection. Each sample is represented as a quadruple $\langle x_{T1}, x_{T2}, x_{txt}, M\rangle$, where (x_{T1}, x_{T2}) is a bitemporal image pair, x_{txt} is a natural-language instruction, and $M \in \{0,1\}^{H\times W}$ is the change mask. For L0 negative (counterfactual) queries, M is an all-zero mask (i.e., an empty mask). ReasonRS-110K contains 112 317 samples in total, and is split into training/validation/testing sets with an 8:1:1 ratio.

To clarify the cognitive demands of our benchmark, we summarize the hierarchical query taxonomy (L0–L3) and its operational criteria in Table II.

1) Human Review and L0 Protocol: To improve the reproducibility and credibility of ReasonRS-110K, we adopt a lightweight quality-control protocol summarized in Table III. Specifically, we manually review a stratified 5% subset of the generated instruction–mask pairs and refine the prompts/filters based on common failure modes (e.g., hallucinated attributes, ambiguous descriptions, or inconsistent change directions). For L0 counterfactual queries (Table II), we construct negatives via semantic mismatch and additionally screen out ambiguous candidates to reduce false negatives, ensuring that compliance metrics (NCA/FPA) reflect genuine abstention behavior. Additional audit details, level-wise pass rates, and error-type statistics are provided in the supplementary material.

To generate high-quality instructions, we use a VLM (Qwen3-VL) to produce mask-conditioned descriptions and then rewrite them into diverse instructions. We further conduct expert verification by randomly sampling 5% of the generated data to filter hallucinated attributes and ambiguous instructions.

B. Evaluation Metrics

We evaluate models in a level-wise manner on ReasonRS-110K (All, L0–L3), following the taxonomy defined in Table II. Since L0 contains negative/counterfactual queries with all-zero masks, we report conventional segmentation metrics on L1–L3 and compliance-oriented metrics on L0.

1) Segmentation Quality (L1–L3): For positive-query levels (L1–L3), we report intersection-over-union (IoU) and $F1$-score:

$$\mathrm{IoU} = \frac{TP}{TP+FP+FN} \tag{13}$$

$$\mathrm{F1} = \frac{2TP}{2TP+FP+FN} \tag{14}$$

where TP, FP, and FN denote the numbers of true positive, false positive, and false negative pixels, respectively. We additionally report the overall score as the weighted average over L1–L3.

2) Negative Compliance (L0): For L0 negative queries, the desired output is an empty (all-zero) mask. Given the predicted probability map $\hat{M}$, we first binarize it with a threshold of 0.5

$$\hat{M}_b = \mathbb{1}\left[\hat{M} > 0.5\right]. \tag{15}$$

We define the normalized predicted area as follows:

$$\mathrm{Area}\left(\hat{M}_b\right) = \frac{1}{HW}\sum_{u,v}\hat{M}_b(u,v). \tag{16}$$

A prediction is considered empty if $\mathrm{Area}(\hat{M}_b) < \tau$, where we set $\tau = 0.001$ (i.e., 0.1% of the image area). We then compute negative compliance accuracy (NCA) as follows:

$$\mathrm{NCA} = \frac{1}{|\mathcal{D}_{L0}|}\sum_{i\in\mathcal{D}_{L0}}\mathbb{1}\left[\mathrm{Area}\left(\hat{M}_b^{(i)}\right) < \tau\right] \tag{17}$$

where $\mathcal{D}_{L0}$ denotes the set of L0 samples. As a complementary measure, we report the false positive area (FPA)

$$\mathrm{FPA} = \frac{1}{|\mathcal{D}_{L0}|}\sum_{i\in\mathcal{D}_{L0}}\mathrm{Area}\left(\hat{M}_b^{(i)}\right) \tag{18}$$

which quantifies how much spurious change area is predicted for negative queries (lower is better).

C. Implementation Details

1) Architecture and Training Environment: Change-LISA is implemented in PyTorch and trained on a single NVIDIA RTX 4090 GPU with 48 GB memory (hardware-modified). SAM ViT-H is used as the Siamese vision backbone ($\mathcal{F}_{enc}$) and LLaVA-v1.5-7B as the multimodal LLM ($\mathcal{F}$). Both $\mathcal{F}_{enc}$ and the main parameters of $\mathcal{F}$ are kept frozen during training. The difference module ($\mathcal{F}_{diff}$), the mask decoder ($\mathcal{F}_{dec}$), and LoRA adapters are optimized. Input images are resized to 1024×1024 with standard data augmentations including random flipping and random cropping.

2) Hyperparameter Settings: AdamW is used with a learning rate of 3×10^{-4} and weight decay of 0. A cosine annealing learning-rate schedule is adopted with a 3% warmup period. The batch size is 2 with gradient accumulation for four steps (effective batch size 8). For parameter-efficient fine-tuning, LoRA adapters are injected into the q_proj and v_proj layers of the LLM with rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.05. The model is trained for ten epochs on the ReasonRS-110K training split.

3) Inference Settings: During inference, we binarize the predicted probability map with a threshold of 0.5 to obtain $\hat{M}$. For negative compliance evaluation, we classify a prediction as an empty mask if the foreground ratio is below $\tau = 0.001$.

D. Comparison Methods

1) Generic CD-Only Models: This group includes representative CNN- and Transformer-based change detection models, namely FC-Siam-Diff [29], SNUNet [55], BIT [7], and ChangeFormer [24], as text-agnostic baselines. For a fair comparison, we retrain all CD-only baselines on ReasonRS-110K by ignoring the instruction x_{txt} and using only the binary change masks as supervision. Since these models do not take language as input, they are optimized to unconditionally segment changes between (x_{T1}, x_{T2}), without selecting regions that satisfy the given instruction. As a result, they can provide competitive performance on positive-query levels (L1–L3), but typically fail on negative/counterfactual queries (L0) under compliance-oriented evaluation.

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2) RIS Models: This group includes state-of-the-art RIS methods, including LAVT [39], GRES [40], RemoteSAM [42], and the original LISA [41]. These methods are designed for single-image, language-conditioned segmentation and do not explicitly model bitemporal changes. To adapt them to our setting, we adopt an early-fusion baseline by channel-wise concatenating x_{T1} and x_{T2} as the visual input while keeping the instruction x_{txt} unchanged; we modify the first projection/convolution layer accordingly to accommodate the concatenated input. For a fair comparison, we use the official implementations of these RIS baselines and fine-tune them on the ReasonRS-110K training split. This group assesses whether language grounding alone can support instruction-following change segmentation, where early fusion may blur temporal ordering and make it difficult to distinguish appeared versus disappeared regions under instruction constraints.

3) Multimodal Large Models: We also compare with MLLMs, including LLaVA-v1.5 and GeoChat [15]. Since these MLLMs do not natively produce pixel-accurate segmentation masks, we adopt a grounding-to-mask pipeline following the Grounded-SAM paradigm. Specifically, we form a side-by-side bitemporal mosaic (left: x_{T1}, right: x_{T2}) as the visual input and prompt the MLLM to output coarse localization cues (bounding boxes) for regions that satisfy the instruction x_{txt}. The predicted boxes are then used as prompts to SAM to generate segmentation masks, which are evaluated using the same metrics as other methods. The mosaic input is used for MLLM baselines to preserve their original RGB input interface and avoid modifying pretrained weights, in contrast to the six-channel early-fusion adaptation used for fine-tuned RIS baselines. Although GeoChat is remote-sensing-oriented, it remains limited to coarse cues in this setting, motivating the necessity of an end-to-end pixel-level decoder for delineating irregular boundaries in remote sensing scenes.

E. Main Results

Table V reports the level-wise performance on ReasonRS-110K. On positive-query levels (L1–L3), Change-LISA achieves 63.1 IoU and 77.3 $F1$. Compared with RemoteSAM [42], the sample-weighted overall score improves by 13.0 IoU and 10.8 $F1$. The method also leads on each positive level (L1:68.5/81.2, L2:62.1/76.5, L3:58.7/73.2).

On L0 negative (counterfactual) queries, Change-LISA achieves 95.4 NCA and 0.12 FPA. This setting evaluates whether a method can abstain by outputting an empty mask when no region matches the instruction. CD-only models yield NCA = 0.00 because they are trained to segment generic temporal differences without language conditioning. GRES [40] attains 72.5 NCA with 1.50 FPA. The improvement of Change-LISA is consistent with combining explicit difference-aware features with an instruction-conditioned decoder, which suppresses spurious masks for negative queries.

F. Cross-Dataset Zero-Shot Evaluation

To evaluate cross-dataset generalization beyond ReasonRS-110K, we further conduct a zero-shot evaluation on SYSU-CD, which is not included in our training data.

Setup: We perform a zero-shot evaluation on SYSU-CD without any additional fine-tuning (see Table VI). SYSU-CD is a large-scale aerial change detection dataset that contains 20 000 image pairs (GSD 0.5 m; 256 × 256) captured in Hong Kong between 2007 and 2014. It covers diverse change types (e.g., buildings, vegetation, roads, and marine structures), which increases the complexity of cross-domain evaluation. Following the official split (6:2:2), we evaluate on the full test split with 4000 bitemporal pairs. For each pair, we construct four instructions following our level-wise taxonomy: one positive instruction for each of L1–L3 (with varied but semantically equivalent phrasing that queries the changed regions) and one L0 negative (counterfactual) instruction that is semantically mismatched to the scene. This results in $4000 \times 4 = 16\,000$ image–instruction queries.

Metrics: We follow the same evaluation protocol as Section IV-B, reporting IoU/$F1$ on positive queries (L1–L3) and NCA/FPA on negative queries (L0), using the same binarization threshold and τ for empty-mask judgment.

Compared Methods: We report results for representative baselines from each category, including a CD-only model (ChangeFormer), a strong RIS baseline (RemoteSAM), and an MLLM pipeline (GeoChat + SAM), together with our Change-LISA.

G. Ablation Study

We conduct ablation studies on ReasonRS-110K to quantify the contribution of key components in Change-LISA. All ablation variants follow the same experimental settings as in Section IV-C. Results are summarized in Table VII, including IoU/$F1$ on positive-query levels (L1–L3) and NCA/FPA on L0 negative queries.

On positive queries (L1–L3), removing the explicit change-aware fusion (Diff/Fusion) degrades IoU/$F1$, indicating that explicit temporal discrepancy modeling improves localization and boundary delineation. On negative queries (L0), excluding negative samples during training leads to a pronounced drop in compliance (lower NCA and higher FPA), showing that counterfactual supervision is essential to suppress hallucinated change masks. Freezing the LLM without LoRA adaptation further reduces both positive performance and compliance, highlighting the importance of parameter-efficient instruction grounding for query-conditioned mask decoding.

1) Ablation on the Design of the Difference/Fusion Module: Beyond the high-level component ablations in Table VII, we further study whether the gains come merely from using a generic difference operator or from the specific design of our $\mathcal{F}_{diff}$. To this end, several variants are compared under the same training and evaluation protocol. These include: 1) subtraction-only (AbsDiff); 2) concatenation-only (Concat); 3) concatenation with explicit AbsDiff (Concat + AbsDiff); and 4) the full design with cross-temporal interaction/alignment followed by AbsDiff and fusion (Align + AbsDiff + Concat). AbsDiff uses $F_{diff} = \phi(|f_{T2} - f_{T1}|)$. Concat uses $F_{diff} = \phi([f_{T1}, f_{T2}])$. Concat + AbsDiff uses $F_{diff} = \phi([f_{T1}, f_{T2}, |f_{T2} - f_{T1}|])$. The full model applies Align$(\cdot)$ before computing $d = |\tilde{f}_{T2} - \tilde{f}_{T1}|$ and fusing $[\tilde{f}_{T1}, \tilde{f}_{T2}, d]$. We report IoU/$F1$ on positive-query levels (L1–L3) and NCA/FPA on L0

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TABLE IX
SENSITIVITY TO LORA RANK. THE DEFAULT SETTING ($r = 8$) IS HIGHLIGHTED

LoRA Rank r	IoU (L1–L3)	F1 (L1–L3)	NCA (L0)↑	FPA (L0)↓
4	62.1	76.5	94.8	0.15
8	**63.1**	**77.3**	**95.4**	**0.12**
16	63.0	77.1	95.2	0.13

TABLE X
IMPACT OF VISION BACKBONE. WE COMPARE DIFFERENT VISION ENCODERS TO VALIDATE THE NECESSITY OF DENSE SPATIAL PRIORS

Encoder	Params	IoU	F1	NCA↑	FPA↓
CLIP-ViT-L [56]	307M	59.8	74.5	92.1	0.45
SAM ViT-B [30]	91M	61.5	76.2	94.2	0.22
SAM ViT-H [30] (Ours)	632M	**63.1**	**77.3**	**95.4**	**0.12**

negative queries in Table VIII. Overall, combining appearance cues with explicit difference (Concat + AbsDiff) improves over single-operator baselines. Adding cross-temporal interaction/alignment yields the best performance (63.1 IoU/77.3 $F1$ on L1–L3 and 95.4 NCA/0.12 FPA on L0). This suggests that alignment and gating help suppress spurious changes under counterfactual queries.

2) LoRA Rank Sensitivity: We analyze the sensitivity of parameter-efficient fine-tuning by varying the LoRA rank r while keeping all other settings unchanged. As shown in Table IX, increasing r from 4 to 8 improves both positive-query segmentation (62.1 → 63.1 IoU; 76.5 → 77.3 $F1$) and negative compliance (94.8 → 95.4 NCA; 0.15 → 0.12 FPA). Further increasing the rank to $r = 16$ brings marginal changes (63.0 IoU, 77.1 $F1$, 95.2 NCA, 0.13 FPA), indicating saturation beyond moderate ranks. Therefore, $r = 8$ is adopted as the default setting to balance accuracy and efficiency.

3) Impact of Vision Backbone: We further study the effect of the vision encoder by replacing the backbone while keeping the rest of Change-LISA and the training protocol unchanged. Table X shows that SAM-based encoders consistently improve both positive-query segmentation and negative compliance. In particular, SAM ViT-H achieves 63.1 IoU and 77.3 $F1$ on L1–L3, outperforming CLIP-ViT-L by 3.3 IoU and 2.8 $F1$. Meanwhile, it also yields higher NCA (95.4) and lower FPA (0.12) on L0, indicating fewer hallucinated change masks under counterfactual instructions. These gains suggest that dense, boundary-sensitive spatial priors learned by SAM are beneficial for producing high-quality change features for subsequent fusion and mask decoding. Notably, CLIP-ViT-L is not a lightweight encoder, yet it remains inferior to SAM variants, implying that the improvement is not solely due to model scale but also to the segmentation-oriented pretraining priors.

4) Sensitivity to Temporal Order: Since the bitemporal inputs (x_{T1}, x_{T2}) are temporally ordered, we analyze whether the model is sensitive to swapping the two timestamps. We randomly sample $N = 50$ test pairs from ReasonRS-110K with direction-dependent instructions (e.g., "newly constructed" versus "demolished") using a fixed seed (seed = 42). We evaluate three settings: 1) normal: (x_{T1}, x_{T2}, q^+); 2) swap-only: (x_{T2}, x_{T1}, q^+), where the original direction-dependent query becomes counterfactual under reversed temporal order and the desired output is an empty mask; and 3) Swap + Flip: (x_{T2}, x_{T1}, q^-), where we flip the instruction semantics (e.g., newly constructed ↔ demolished) to restore temporal consistency. Following Section IV-B, we report IoU/$F1$ for 1) and 3), and NCA/FPA for the Swap-only setting 2) using the same empty-mask threshold τ. Table XI summarizes the results: Swap + Flip remains close to Normal (66.8/80.1 versus 65.5/79.2 in IoU/$F1$), while Swap-only achieves 93.4 NCA with 0.22 FPA. Fig. 5 provides a representative qualitative example.

TABLE XI
TEMPORAL ORDER SENSITIVITY ON REASONRS-110K. WE SAMPLE $N = 50$ DIRECTION-DEPENDENT TEST PAIRS (SEED = 42). IOU/$F1$ ARE REPORTED FOR TEMPORALLY CONSISTENT SETTINGS, WHILE NCA/FPA EVALUATE THE COUNTERFACTUAL SWAP-ONLY SETTING WHERE THE DESIRED OUTPUT IS AN EMPTY MASK

Setting	IoU	F1	NCA↑	FPA↓
Normal $(x_{T1} \to x_{T2}, q^+)$	66.8	80.1	–	–
Swap-only $(x_{T2} \to x_{T1}, q^+)$	–	–	93.4	0.22
Swap+Flip $(x_{T2} \to x_{T1}, q^-)$	65.5	79.2	–	–

H. Qualitative Analysis

Fig. 6 presents qualitative comparisons on ReasonRS-110K. The cases highlight query-conditioned selectivity, i.e., differentiating target changes from irrelevant ones based on the instruction. Negative compliance under counterfactual queries is evaluated separately with NCA/FPA (Section IV-B) and illustrated in Figs. 1 and 5.

Across the cases, the CD-only baseline ChangeFormer tends to produce over-inclusive masks because it is optimized to capture generic temporal differences regardless of the query. This often merges target regions with surrounding nontarget areas. RemoteSAM, without explicit bitemporal change modeling, can be biased by semantic cues and yields noticeable false positives in challenging scenarios.

In contrast, Change-LISA produces predictions that better match the query intent. Across cases with different prompts (e.g., water-body changes, deforestation, new construction, and facility construction), it delineates the queried change regions with cleaner boundaries and fewer spurious responses than the baselines.

I. Failure Case Analysis

To further analyze the limitations of Change-LISA, Fig. 7 presents representative failure cases under text-guided change segmentation. Although the text queries provide explicit semantic guidance, the model can still produce incorrect predictions in challenging scenes. The observed errors mainly include false positives over visually similar background regions, missed detections of subtle target changes, and imprecise boundaries around irregular structures.

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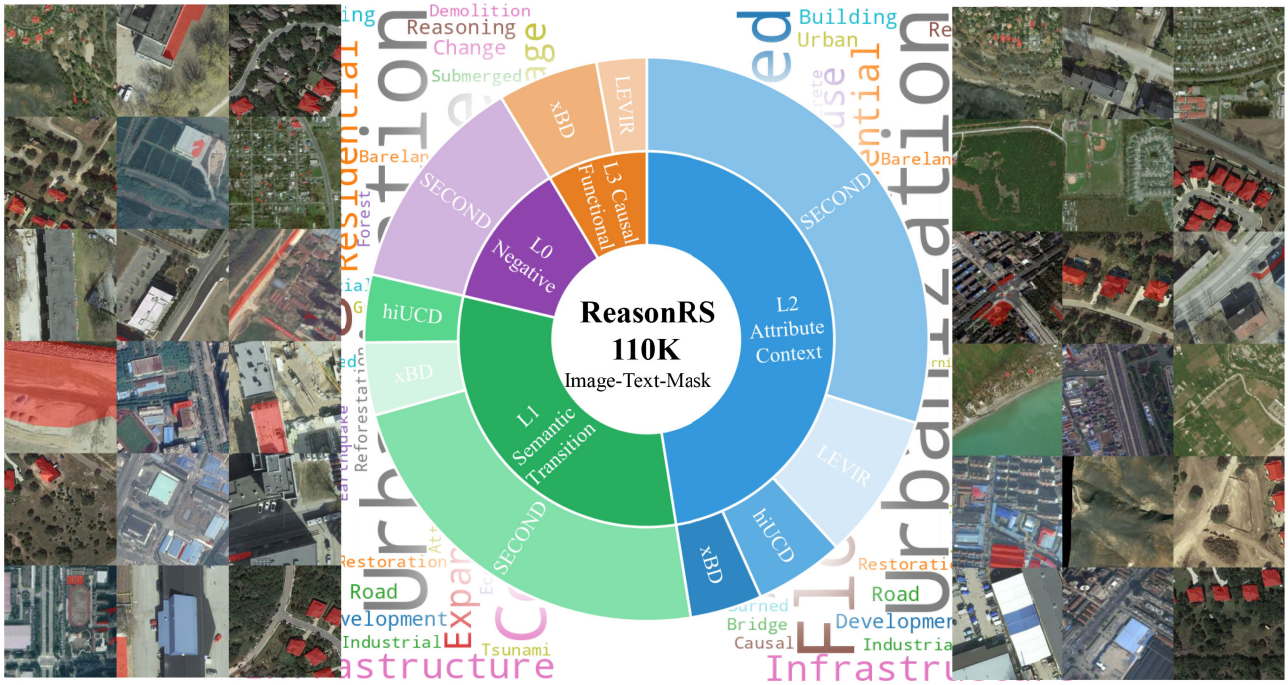


Fig. 4. Overview of the proposed ReasonRS-110K benchmark. The dataset is constructed through multisource integration (SECOND [49], xBD [50], LEVIR [10], and Hi-UCD [51]) to cover diverse remote sensing scenarios. Center: the sunburst chart illustrates the hierarchical data distribution, consisting of four reasoning levels: L0 negative (counterfactuals), L1 semantic transition (basic land-cover changes), L2 attribute context (fine-grained urban attributes), and L3 causal functional (disaster and infrastructure reasoning). Background: the word cloud visualizes the rich vocabulary of our instruction set. Sides: sample image-mask pairs demonstrate the high diversity of scenes and change types.

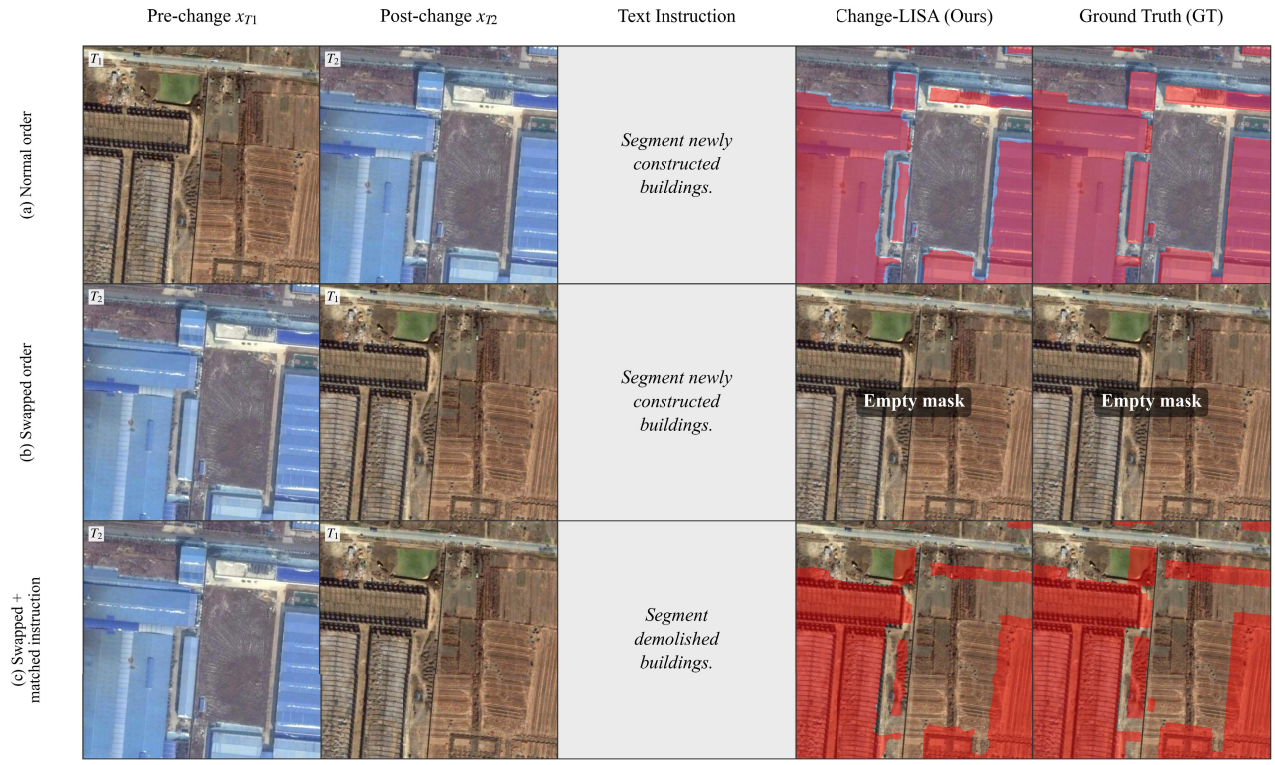


Fig. 5. Qualitative example of temporal order sensitivity. (a) Normal order ($x_{T1} \rightarrow x_{T2}$) with a direction-dependent instruction ("newly constructed"), where the prediction matches the GT. (b) Swaps the timestamps while keeping the instruction unchanged, which makes the query counterfactual under the reversed temporal order and leads to an empty mask. (c) Swaps the timestamps and flips the instruction semantics ("newly constructed" ↔ "demolished"), restoring temporal consistency and yielding a correct prediction.

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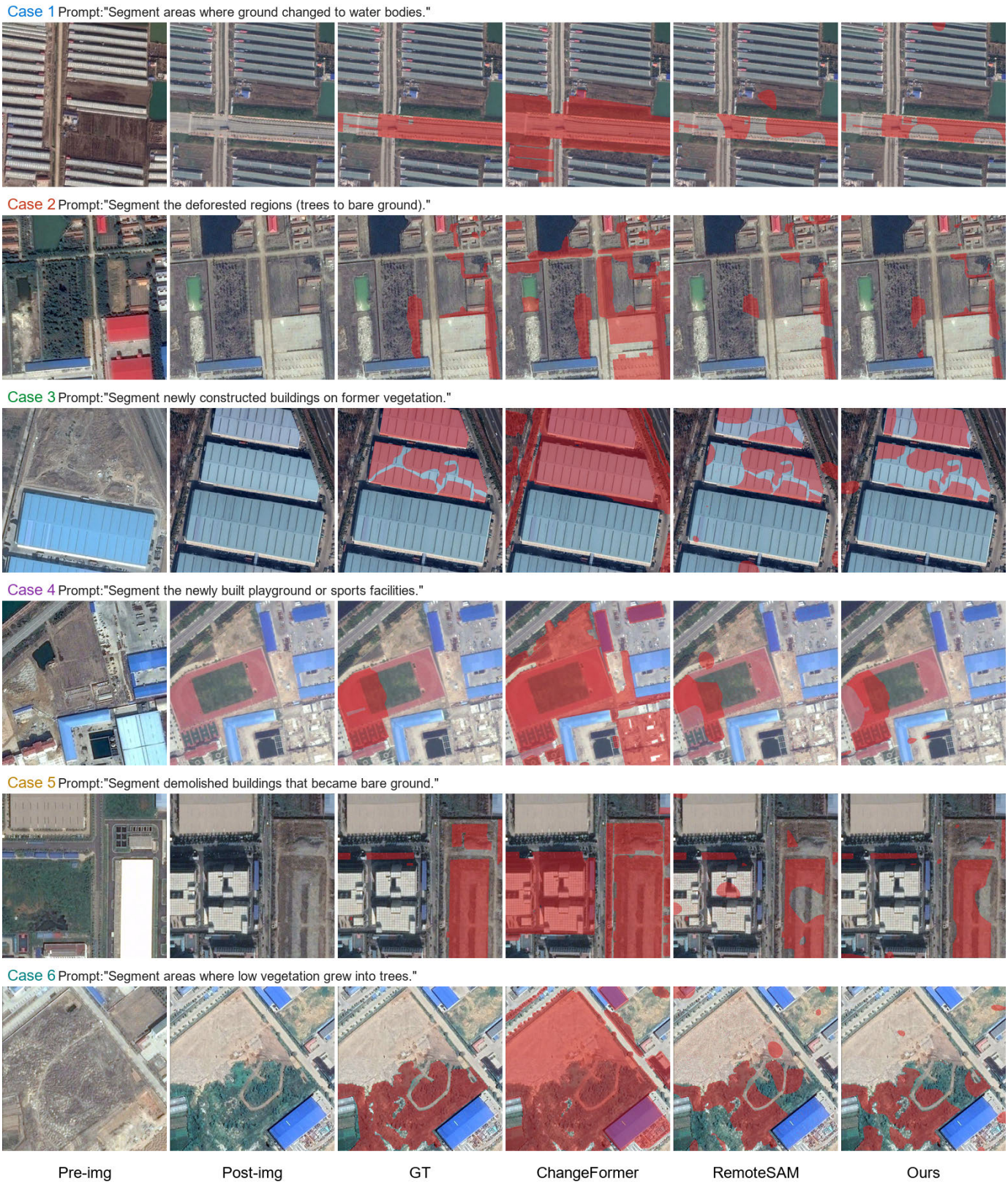


Fig. 6. Qualitative comparison on ReasonRS-110K. We show six cases with different prompts. (Left to right) Prechange image, postchange image, GT, predictions of hangeFormer [24], RemoteSAM [42], and our Change-LISA. Red overlays indicate the predicted change masks.

These cases indicate that text guidance alone is not always sufficient to fully resolve semantic ambiguity in remote sensing imagery. Complex land-cover patterns, background clutter, and subtle appearance variations between bitemporal images may still mislead the model and weaken the alignment between the query and the true change evidence. Future improvements may therefore benefit from stronger cross-modal alignment and finer spatial discrimination.

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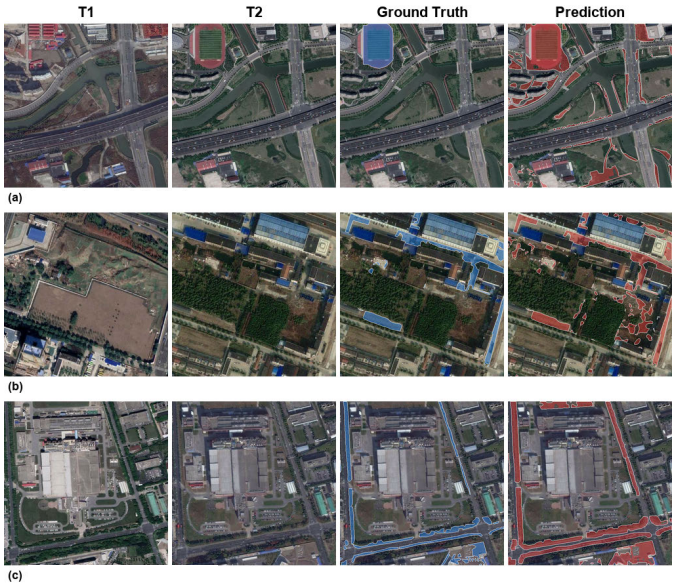


Fig. 7. Representative failure cases of text-guided change segmentation. From left to right are the prechange image (T1), postchange image (T2), GT mask, and prediction. Each example is accompanied by its text query. Blue and red overlays denote the GT and predicted changed regions, respectively. Although the query provides explicit semantic guidance, the model still yields incorrect segmentation results in complex scenes. (a) Query: "newly constructed buildings." (b) Query: "areas where trees were removed." (c) Query: "vegetation cleared to bare ground."

V. Conclusion

In this article, we studied language-guided reasoning change detection for remote sensing and introduced the ReasonRS-110K benchmark with hierarchical reasoning levels (L1–L3) and counterfactual negative queries (L0) for compliance evaluation. We proposed Change-LISA, a parameter-efficient framework integrating a frozen Siamese SAM encoder, an explicit difference/fusion module, and a multimodal LLM (LLaVA) to prompt a pixel-level mask decoder via a <SEG> token. On ReasonRS-110K, Change-LISA achieves 63.1 IoU and 77.3 $F1$ on L1–L3 and 95.4 NCA with 0.12 FPA on L0. Ablation results indicate that explicit temporal discrepancy modeling, negative-query training, and LoRA-based LLM adaptation each contribute to accuracy and compliance. A limitation is that the framework still depends on the behavior of the underlying LLM and the instruction set; future work will study broader generalization across sensors/regions and more efficient architectures for deployment.

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